Chapter — Major Errors Encountered During Development

During the development of the Retail AI Analytics Platform, multiple technical issues occurred while configuring the environment and implementing the analytics pipeline. This chapter documents the most significant errors and how they were resolved.

Error 1 — Spark Not Starting (JAVA_GATEWAY_EXITED)

Error Message

PySparkRuntimeError: JAVA_GATEWAY_EXITED

Cause

Spark could not start because Java was either not installed correctly or the JAVA_HOME environment variable was missing.

Solution

Java Development Kit (JDK) version 17 was installed and the JAVA_HOME variable was configured correctly.

Example configuration:

JAVA_HOME = C:\Program Files\Eclipse Adoptium\jdk-17.0.18.8-hotspot

After this configuration Spark started successfully.

Error 2 — Java Version Compatibility

Error

Spark failed to run when using a newer Java version.

Cause

Some Spark builds are not compatible with the newest Java versions.

Solution

Java 17 was installed because it is widely supported by Apache Spark.

After switching to Java 17 the compatibility issue disappeared.

Error 3 — Python Version Conflict

Issue

Spark sometimes executed with the wrong Python interpreter.

Cause

Multiple Python versions were installed on the system.

Solution

The project explicitly used Python 3.11 by running commands like:

py -3.11 script_name.py

Example:

py -3.11 spark_jobs/process_sales.py

This ensured Spark always used the correct interpreter.

Error 4 — Spark SQL Column Not Found

Error

UNRESOLVED_COLUMN error for "SALES".

Cause

The dataset did not contain a column called SALES.

Solution

A calculated column was created using:

sales = Quantity * UnitPrice

This column was then used in Spark SQL aggregation queries.

Error 5 — Negative Machine Learning Predictions

Issue

The machine learning model occasionally produced negative sales predictions.

Example:

Estimated Sales Value: $-1942.66

Cause

Regression models can produce negative values when extrapolating beyond the training data.

Solution

A constraint was added to ensure predictions remain valid.

Example:

prediction = max(prediction, 0)

Error 6 — Spark Data Not Displaying in Streamlit

Issue

Spark DataFrames did not display correctly in the dashboard.

Cause

Streamlit works best with Pandas DataFrames.

Solution

Spark DataFrames were converted to Pandas before visualization.

Example:

df_pandas = spark_df.toPandas()

Error 7 — Dashboard Layout Problems

Issue

Some charts overlapped or appeared misaligned in the dashboard.

Cause

Improper layout structure in Streamlit.

Solution

The layout was redesigned using column structures.

Example:

col1, col2 = st.columns(2)

This created a cleaner and more balanced dashboard layout.

Error 8 — Real-Time Data Not Updating

Issue

The live sales monitor did not update automatically.

Cause

The dashboard needed a refresh mechanism.

Solution

Auto-refresh logic was implemented so the dashboard reloads new data periodically.

Error 9 — Git Upload Confusion

Issue

There was uncertainty about whether a new GitHub repository needed to be created.

Cause

The repository already existed but updates had not been pushed.

Solution

Git commands were used to push updates:

```
git add .
git commit -m "Project updates"
git push
```

Error 10 — Dashboard UI Quality Improvements

Issue

Initial charts and metrics were functional but not visually polished.

Solution

UI improvements were added:

• KPI metric cards
• interactive filters
• improved map visualization
• better chart layout

This significantly improved dashboard usability.

Key Learning From Debugging

These errors helped develop several important skills:

• reading and interpreting stack traces
• configuring development environments
• understanding Spark dependencies
• improving dashboard design
• debugging machine learning predictions

Debugging and solving these issues was a critical part of completing the project.

Chapter — End-to-End System Explanation

This chapter explains how the complete Retail AI Analytics Platform works from raw data ingestion to the final analytics dashboard.

The system simulates a real business analytics platform used by companies to monitor sales data and generate insights.

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1. Data Source

The project begins with a retail dataset containing transactional sales data.

Example fields include:

InvoiceNo
StockCode
Description
Quantity
InvoiceDate
UnitPrice
CustomerID
Country

Each row represents a transaction in the retail system.

This dataset acts as the raw input for the analytics pipeline.

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2. Big Data Processing

The raw dataset is processed using Apache Spark.

Spark is used because it is designed for distributed big data processing.

Key processing tasks include:

• loading the CSV dataset
• cleaning and validating data
• computing derived columns
• aggregating sales statistics

One important computed column is:

sales = Quantity × UnitPrice

This value represents the total value of each transaction.

Spark then aggregates the dataset to compute insights such as:

• total sales per country
• product popularity
• global revenue distribution

These results are saved into processed datasets used by the dashboard.

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3. Machine Learning Prediction

A machine learning model is used to estimate potential sales values.

The model is trained using historical sales information such as:

Quantity
UnitPrice
Country

A regression model from the Scikit-learn library is used for prediction.

The model estimates future sales values based on user input.

To ensure realistic predictions, a validation step ensures that predicted values cannot be negative.

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4. Real-Time Sales Simulation

To simulate live business activity, a Python script generates random sales transactions.

The script runs continuously and produces records such as:

timestamp
country
product
quantity
price

These records are appended to a dataset used by the dashboard.

This simulates a real-time streaming system similar to production analytics pipelines.

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5. Interactive Analytics Dashboard

The processed data is visualized using an interactive dashboard built with Streamlit.

The dashboard provides several pages:

Overview
Displays key performance indicators including revenue and order metrics.

Sales Analytics
Provides charts showing sales distribution across countries and products.

ML Predictions
Allows users to input values and generate predicted sales estimates.

Live Sales Monitor
Displays simulated real-time transactions and dynamic charts.

Business Reports
Allows users to generate downloadable reports in PDF or Excel format.

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6. Business Insights

The platform helps answer important business questions such as:

Which countries generate the highest revenue?

Which products are most popular?

How is revenue distributed globally?

What is the predicted sales value for a given order?

These insights can help companies make better business decisions.

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7. System Workflow Summary

The full pipeline can be summarized as:

Raw Retail Data
→ Spark Data Processing
→ Aggregated Analytics Data
→ Machine Learning Prediction
→ Interactive Dashboard Visualization

This structure reflects the architecture used in real data analytics systems.

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8. Key Skills Developed

Building this project required knowledge in several areas:

Big data processing
Machine learning
data visualization
system debugging
dashboard design

The project demonstrates how these technologies can be combined to build a complete analytics platform.